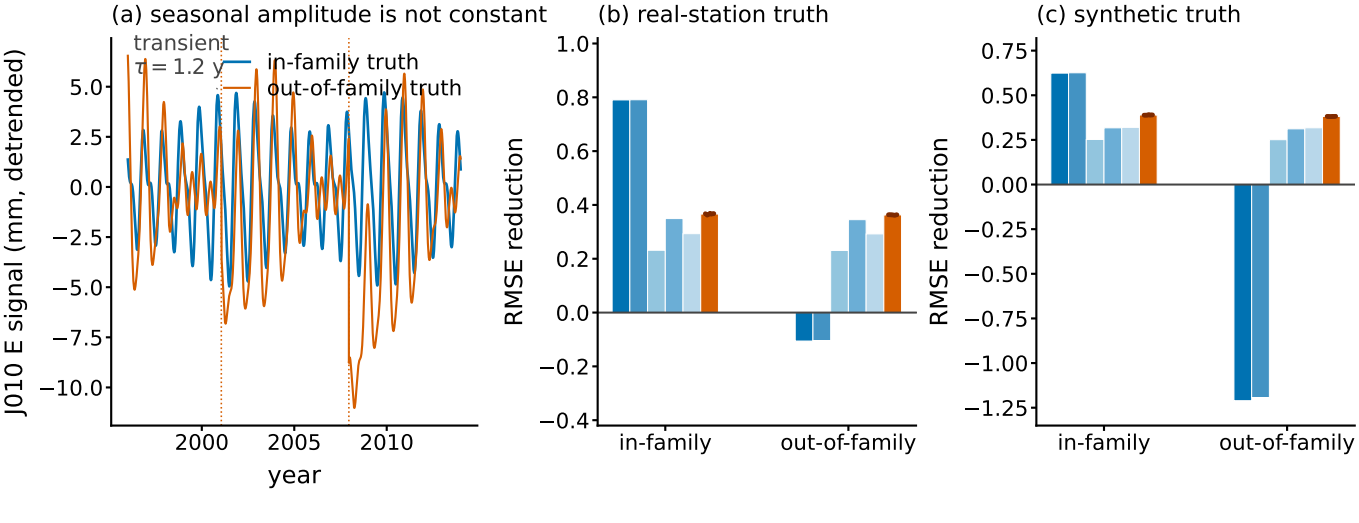
(a) seasonal amplitude is not constant
transient
τ = 1.2 y
in-family truth
out-of-family truth
J010 E signal (mm, detrended)
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
2000
2005
2010
year
(b) real-station truth
RMSE reduction
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
in-family
out-of-family
(c) synthetic truth
RMSE reduction
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
in-family
out-of-family
MLE
RTS
SSA
wavelet
median
U-Net